

House Price Prediction with Generalized Linear Model (GLM)

Project Overview

In this project, a Generalized Linear Model (GLM) was developed to estimate house prices using selected housing characteristics from the Kaggle House Prices dataset.

The main objective of the project is to examine the relationship between house prices and structural housing features while evaluating the predictive performance of the GLM approach.

In order to create a more interpretable and simplified model, only selected explanatory variables were used instead of the full dataset.

Dataset

The dataset used in this project is the "House Prices - Advanced Regression Techniques" dataset obtained from Kaggle.

Initially, the dataset contained:

- 80 independent variables
- 1 target variable (SalePrice)

To build a more interpretable model, the following four explanatory variables were selected:

- GrLivArea → Above ground living area
- OverallQual → Overall material and finish quality
- YearBuilt → Construction year
- TotalBsmtSF → Total basement area

Target Variable:

- SalePrice → House sale price
-

Data Preprocessing

The dataset was first examined for missing observations.

Missing values were removed using the `dropna()` method.

Additionally, logarithmic transformation was applied to the target variable:

```
df["SalePrice"] = np.log(df["SalePrice"])
```

The purpose of this transformation was:

- to reduce skewness in house prices,
- stabilize variance,
- minimize the effect of extreme values,

- and improve model performance.

Applying log transformation allowed the model to produce more stable and interpretable results.

Train-Test Split

To evaluate model performance properly, the dataset was divided into training and test sets.

- Training data: 80%
- Test data: 20%

This approach allowed the model to learn from one portion of the dataset and then be evaluated on previously unseen observations.

The train-test split process is important for understanding whether the model generalizes well or simply memorizes the training data.

Model Construction

The project was developed using Python programming language.

A Generalized Linear Model (GLM) with Gaussian family was constructed using the statsmodels library.

The model equation is based on:

$\text{SalePrice} \sim \text{GrLivArea} + \text{OverallQual} + \text{YearBuilt} + \text{TotalBsmtSF}$

The GLM approach was preferred because of its interpretability and ability to clearly show the relationship between explanatory variables and house prices.

Model Results

The following performance metrics were obtained from the model:

- Pseudo R^2 : 0.9734
- Mean Absolute Error (MAE): 0.1227

The pseudo R^2 value of approximately 0.97 indicates that the selected variables explain a very large proportion of the variation in house prices.

This result suggests that the model has a strong fit to the dataset.

Model performance was also evaluated using Mean Absolute Error (MAE).

The MAE value of approximately 0.12 indicates that prediction errors remain relatively low.

Since logarithmic transformation was applied to the target variable, the MAE value should be interpreted on the logarithmic scale.

Overall, the low MAE value suggests that the model generated relatively accurate predictions.

Interpretation of Coefficients

All variables were found to be statistically significant according to the model output.

For all variables:

- $p\text{-value} < 0.05$

was obtained.

This indicates that all selected explanatory variables have statistically meaningful effects on house prices.

Coefficient analysis showed that:

- OverallQual has the strongest positive impact on house prices.
- Larger living areas are associated with higher prices.
- Newer houses tend to have higher market values.
- Basement area also contributes positively to house prices.

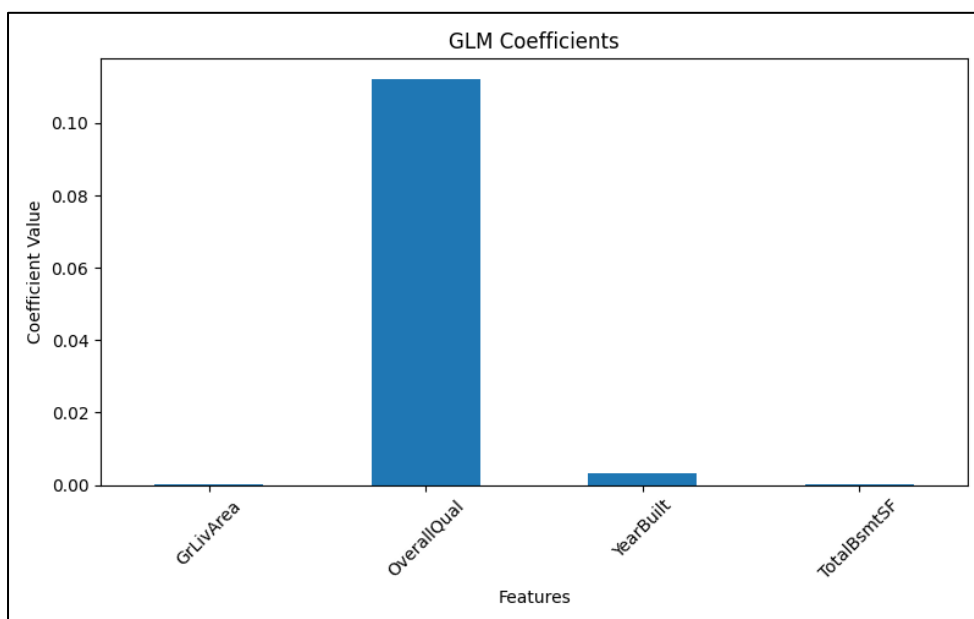


Figure 1: Comparison of GLM coefficient values for selected housing variables.

The figure below presents the coefficient values of the selected variables.

Conclusion

The results demonstrate that the Generalized Linear Model (GLM) approach can successfully estimate house prices using a limited number of explanatory variables.

Throughout this project, several important data science and statistical modeling processes were applied, including:

- data preprocessing,
- missing value handling,
- logarithmic transformation,
- train-test split,
- regression modeling,
- model evaluation,
- and coefficient interpretation.

Overall, the model produced interpretable and successful prediction results while demonstrating strong explanatory performance.